Replicating FinBERT

An Independent Research Project in Financial NLP & Transfer Learning

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Project Summary

This project is an independent replication and extension of a published machine learning thesis: “FinBERT: Financial Sentiment Analysis with Pre-trained Language Models” (Araci, 2019, University of Amsterdam) . Building on the original paper, I modified a BERT transformer model to classify sentences from financial news as positive, negative, or neutral. I compared my results with those reported in the paper. I also created two new extensions, a full-dataset evaluation and an interpretability analysis on attention levels. This allowed me to assess the robustness of the paper’s results under different conditions. Overall, this project was about building a complete Natural Language Processing (NLP) pipeline using PyTorch and the Hugging Face ‘transformers’ library. More specifically, I trained the model on Google Colab, and also compared my results to those reported in the source paper itself rather than just reproducing a single reported number.

Background & Motivation

Financial sentiment analysis, automatically determining if a piece of financial text is good or bad news for a company, is a really hard NLP problem because financial language is specialized, understated and often based on numeric comparisons rather than obviously positive or negative words. In the original FinBERT paper this was solved by taking BERT, a large pre-trained language model, and fine-tuning it on a labeled set of financial sentences (the Financial PhraseBank). I decided to replicate this paper because it combines two things I wanted to learn by doing: how modern transformer-based language models actually work under the hood, and how quantitative methods get applied to real financial decision-making.

Methodology

Dataset

Financial PhraseBank (Malo et al., 2014): English sentences from financial news, labeled positive, negative, or neutral by 16 finance professionals based on how the news would likely affect the mentioned company's stock price. I evaluated on two versions of this dataset to test how much the reported results depend on dataset selection:

- 100%-Agreement Subset (2,262 sentences) : only sentences where all annotators agreed on the label
- Full Dataset (4,845 sentences) : includes lower-agreement, more ambiguous sentences

Model & Tools

- Base model: bert-base-uncased (Hugging Face Transformers) fine-tuned with a custom classification head
- Framework: PyTorch, Hugging Face Transformers, scikit-learn
- Training environment: Google Colab (T4 GPU)
- Development assistance: Claude Code (pipeline development & debugging), Google Gemini (research support)

Hyperparameters

Parameter	Value
Max sequence length	64 tokens
Batch size	32
Optimizer	AdamW
Learning rate	2e-5
Adam epsilon	1e-8
Gradient clipping	max_norm = 1.0
Epochs	3

Results

100%-Agreement Subset

Class	Precision	Recall	F1-score	Support
Negative	0.92	0.96	0.94	56
Neutral	0.99	0.99	0.99	276
Positive	0.97	0.94	0.95	121
Overall accuracy			97.35%	453

Full Dataset

Class	Precision	Recall	F1-score	Support
Negative	0.70	0.88	0.78	121
Neutral	0.90	0.82	0.86	576
Positive	0.74	0.79	0.77	273
Overall accuracy			82.16%	970

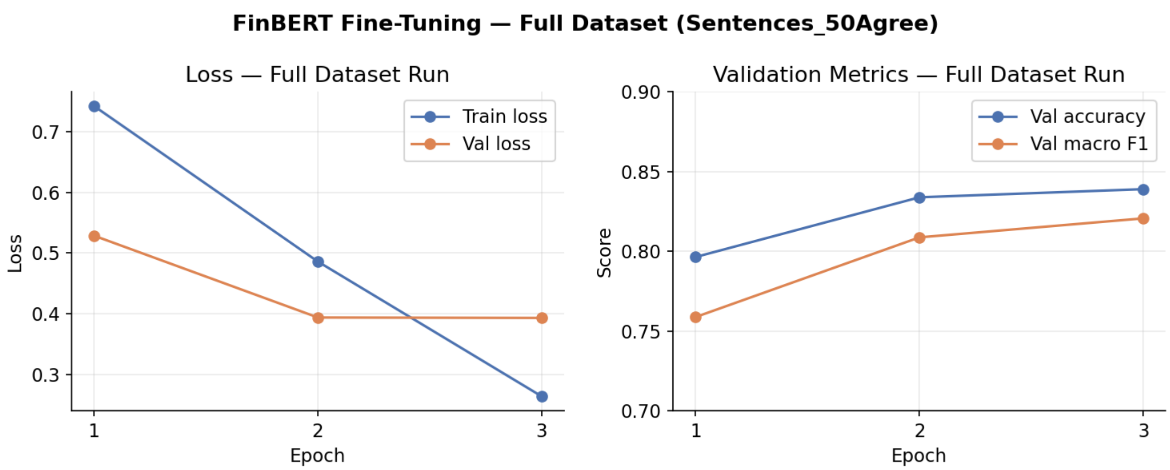

Figure 1. Training and validation loss/metrics across 3 fine-tuning epochs (full dataset run).

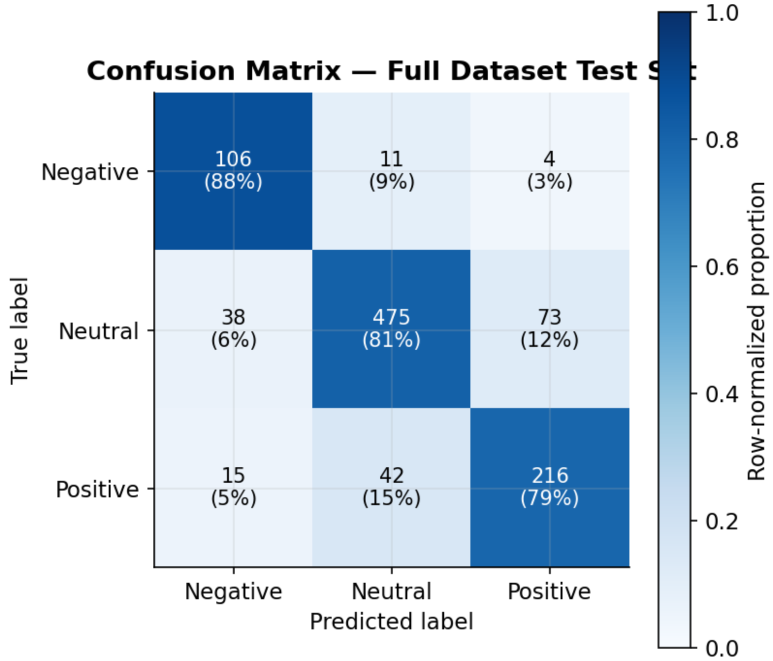

Figure 2. Confusion matrix on the full-dataset held-out test set (970 examples).

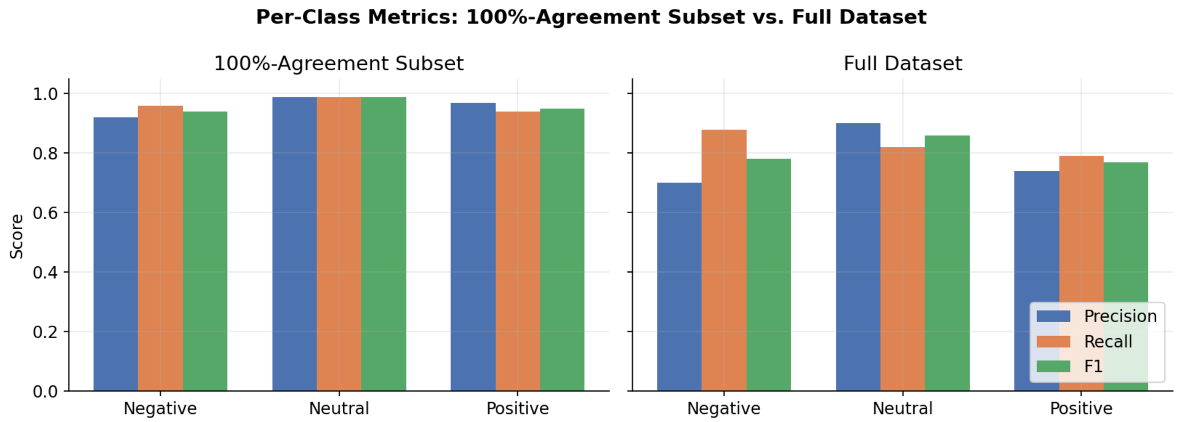

Figure 3. Per-class precision/recall/F1: 100%-agreement subset vs. full dataset.

Conclusion & Discussion

Rather than stopping at a single headline accuracy number, I compared my results directly against the paper's reported numbers across both dataset conditions, which showed a finding I did not expect.

Model	Dataset	Accuracy	F1
Paper - FinBERT (full pipeline)	100% agreement	0.97	0.95
Paper - FinBERT (full pipeline)	Full data	0.86	0.84
Paper - Vanilla BERT (ablation)	Full data	0.85	0.84
My model	100% agreement	0.97	0.96
My model	Full data	0.82	0.80

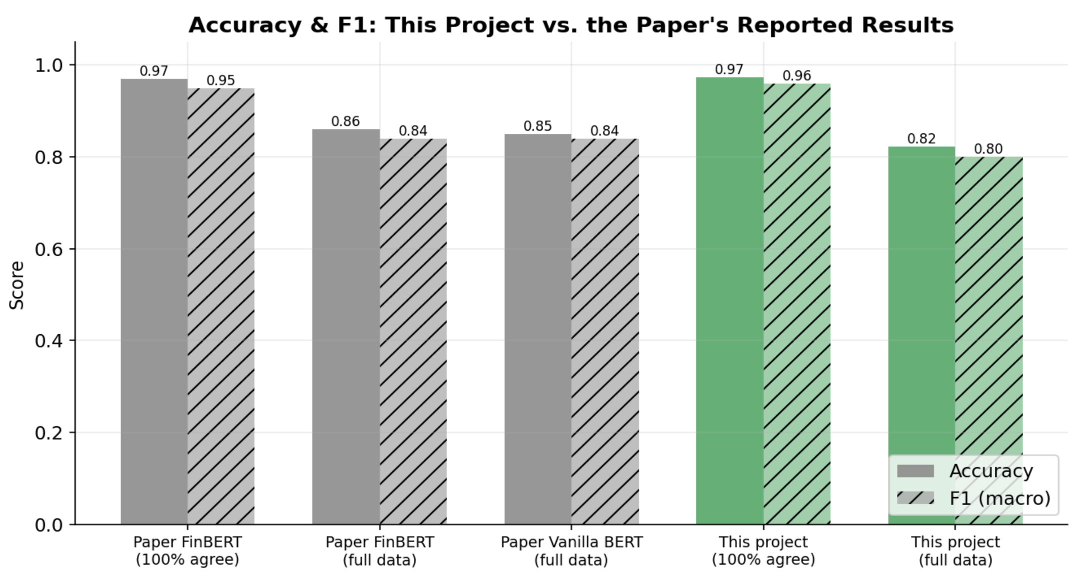

Figure 4. My results vs. the paper's reported results, across both dataset conditions.

While this is very close to the result that the paper reported for the FinBERT system, this result is still potentially misleading as a headline result. The percentage of correct predictions here is based only on the “easiest” subset of the data set where all of the annotators agreed on the label for each news article. When I re-applied the same model to the entire, more difficult data set, I dropped to an accuracy of 82.16%. This is worse than the Vanilla BERT model that was presented by the paper (which obtains 85% accuracy on the same data set). The omitted training techniques intended to prevent the model from “forgetting” what it had learned during pre-training is one reason why the paper’s model performs better. This kind of result is not something that is seen until testing the claims of the paper on data sets that are not covered by its headline result.

Interpretability: Where Does the Model Fail?

I built an attention-visualization and gradient-based saliency tool to try to understand not only how accurate the model was, but why it made specific mistakes, and tested it on the three exact failure examples the original paper discusses in its own error analysis.

Example	True Label	My Model's Prediction	Result
“Pre-tax loss totaled €0.3M vs. €2.2M loss...”	Positive	Negative	Same failure as the paper
“Fixed to Mobile convergence launch in Brazil...”	Neutral	Neutral	Correct, paper's model failed this one
“Coated magazine printing paper... continue to be weak”	Negative	Negative	Correct, paper's model failed this one

My model replicated the paper's exact numeric-reasoning issue in the first example. It failed to see that a smaller loss represents an improvement, probably because BERT's subword tokenizer breaks numbers like “0.3” and “2.2” into parts that lack any sense of scale. Meanwhile, it successfully handled two examples that needed a clear separation between a simple fact and a real sentiment signal. The paper's model misjudged this distinction. This provides a useful, specific insight rather than a vague conclusion that “the model works well.” Transformer sentiment models excel at identifying qualitative tone but struggle with quantitative comparison.

Skills Demonstrated

- Applied machine learning: fine-tuning a pre-trained transformer (BERT) for a real classification task
- Python, PyTorch, and the Hugging Face Transformers ecosystem
- Experimental methodology: train/validation/test splitting, checkpoint selection, stratified sampling
- Critical evaluation of published research: identifying where a paper's headline result does and doesn't generalize
- Model interpretability: attention visualization and gradient-based saliency analysis
- Data visualization: building publication-quality charts to communicate quantitative findings
- Technical writing and documentation: maintaining a clear, versioned project README throughout

Reflections

The most valuable part of this project wasn't getting a high accuracy number, it was learning not to just trust my first set of results. I started with 97% accuracy, which looked great, but when I looked closer at the original paper's methods, I realized the test was much narrower than I thought. I decided to run the experiment again using more difficult and realistic conditions, which gave me a much more honest look at what the model could and couldn't do. I found that questioning a satisfactory result instead of just accepting it was the most important part of the whole research process as it helped me go in depth in the topic and learn more.

References & Resources

Araci, D. (2019). FinBERT: Financial Sentiment Analysis with Pre-trained Language Models. arXiv:1908.10063.

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